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Green Solution For Sacred Waste: Developing Eco-Friendly Aromatic Product

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Abstract

The great cultural and spiritual heritage of India produces a tremendous amount of floral waste, mostly from temples. Though the wastes are sacred in nature, proper management is hardly done, thereby causing problems for the environment by polluting water and producing bad gases at the time of its decomposition[1]. For this reason, the current research work has tried to transform temple flower waste into eco-friendly dhoops and incense sticks, thus motivating sustainable development with a circular economy.[2]

This project included the collection of flower waste, preprocessing it into usable raw materials, and formulating dhoops and incense sticks using natural binders and essential oils.[3] Different formulations were developed to improve binding strength, burning efficiency, and fragrance while eliminating the use of carbonaceous materials like camphor and charcoal.[4] Natural ingredients like gum benzoin, gugal, and babool gum improved product performance and contributed to eco-friendliness by reducing emissions and ash residue.

Results will show that products can be produced from temple flower waste successfully. This means practical waste management and the resultant product is also an environmentally conscious alternative to conventional products.[5] The results of this study can be used to scale up such initiatives and benefit communities by promoting environmental conservation through innovative and green practices.[6]

Key Words : Flower waste management, Sustainable products, temple waste reuse, Dhoop sticks.

1. Introduction

1.1 Background and Significance

India, known for its vibrant festivals and deep-rooted spiritual traditions, produces a large volume of flower waste from temples.[7] These discarded flowers, often considered sacred, are disposed of inadequately in water bodies or abandoned in open areas, resulting in environmental contamination.[8] When this waste decomposes without oxygen, it releases harmful gases such as methane and ammonia, and it also pollutes waterways when

released into rivers. Furthermore, the use of pesticides and fertilizers in flower farming further diminishes the quality of these water bodies.[9]

1.2 The Importance of Sustainability and Green Products

Implementing sustainable practices is essential for safeguarding the environment. Transforming temple flower waste into valuable products fosters a circular economy and lessens the reliance on new raw materials. Eco-friendly products created from recycled materials are becoming increasingly popular due to their sustainable attributes.[10], [11]

1.3. Objectives

The goals of this project are to:

- Tackle the problem of temple flower waste by developing environmentally friendly products from this waste stream.
- Encourage sustainable practices and offer waste management solutions.
- Introduce the idea of a circular economy within the temple community.

2. Methods and Materials

2.1 Material Sourcing

Waste flowers were procured from Siddhivinayak Temple at Prabhadevi, Mumbai. Paraffin wax was purchased from G Kuntala and Company. Cow dung powder was procured from a cow shelter in Vadala. Palm oil and Mahua crude oil were purchased from the Asha Oil depot in Matunga. Gum benzoin, Gugal, Jigat powder, Gond powder, Xanthan Gum, Babool Gum, and Gandh biroza were purchased from the Acharya and Ajanta Agarbatti store.

2.2 Preprocessing

It took around 1 hour for the petals to dry completely. The dried flower waste was ground into a fine powder using a mechanical grinder. Any impurities, such as non-organic material or debris, were removed through sieving to ensure uniformity. The flower powder was then mixed with the appropriate material to make the final formulation. The collected waste flower petals were dried in a laboratory oven at 80 degrees Celsius.

2.3 Formulation process

The flower powder was mixed with cow dung powder (a natural binder that provides cohesiveness), camphor, and palm oil. The mixture was blended in a mixture to get a uniform consistency. The natural essential oils of the white udh, sandalwood, and jasmine types were incorporated into the mixture. The proportion of essential oils was adjusted to ensure that there was a good, pleasant odor without overpowering the inherent odor of the base material. The mixture was gradually moistened with a small quantity of water to achieve a dough-like consistency. Kneading ensured the binder and essential oils were uniformly distributed. The kneaded mixture was allowed to set for 30 minutes. The prepared mixture was pressed into a custom-designed steel mold to achieve uniform dhoop sticks. They were then sun-dried for 24-48 hours. Controlled drying prevented cracking and ensured a smooth finish.

2.4 Mold design

To ensure uniformity and consistency in the production of dhoop sticks, a custom-designed mold was fabricated. The mold was specifically tailored to accommodate the requirements of the sustainable material and facilitate ease of use during the manufacturing process.

Figure No. 2.1 - This figure illustrates the sectional view of the mold.

Figure No. 2.2 - This figure illustrates the top view of the mold.

The mold was constructed using stainless steel due to its durability, resistance to corrosion, and ability to withstand repeated use. The smooth surface of the mold made demolding easy without causing damage to the dhoop sticks. [Fig 2.1] [Fig 2.2]

2.4.1 Design specifications

A vertical cavity was designed to produce dhoops with constant dimensions and homogenous density. It has screws or alignment holes to hold the mold halves in place during filling and pressing.

2.4.2 Functionality

The mold was designed to handle semi-solid formulations under sufficient pressure for compactness. The cavity size was optimized for efficient drying and combustion so that the final product would have uniform burning characteristics.

2.4.3 Ease of operation

The mold can be filled, compacted, and de-molded in a short time, which improves production efficiency. Its design makes it easy to clean and maintain after repeated use.

2.4.4 Sustainability aspect

The use of a reusable mold is aligned with the sustainability goal of the project since material waste during production is minimized.

2.5 Performance assessment

This section emphasizes the parameters, testing methods, and results that evaluate the performance and eco-friendliness of dhoops and incense sticks from temple flower waste.

2.5.1 Fragrance Retention

Products were tested for 0 - 30 days of storage for retention. The samples are placed in air-tight containers under normal conditions for 30 days. Periodically, the gas containers are opened, and then with the help of a GC-MS (gas chromatography-mass spectrometry) setup [Fig 2.3], the volatility of the volatile compounds is analyzed to ensure that the essential oils remain potent. Based on the results, the strength of the aroma was evaluated.

Figure No. 2.3 - This figure illustrates the GC-MS (gas chromatography-mass spectrometry) set-up to analyse volatile compounds.

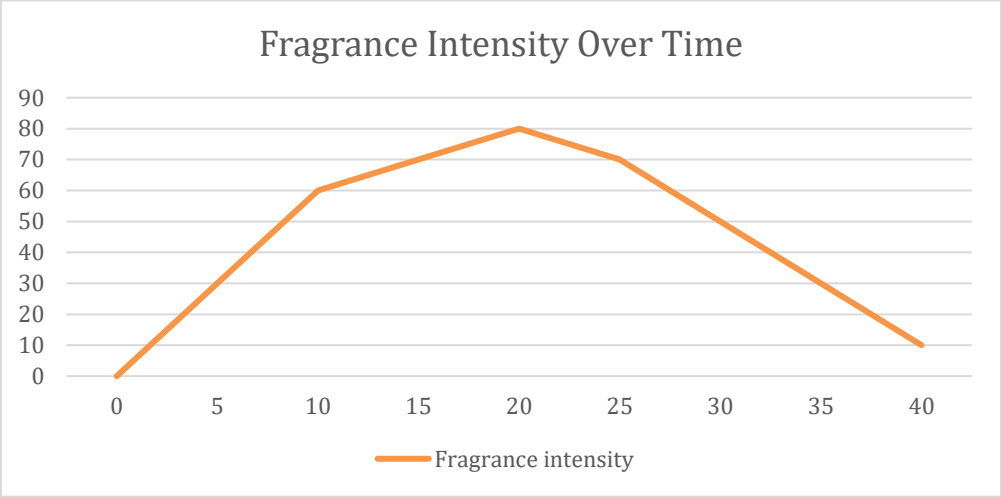

Figure No. 2.4 - The figure illustrates that the fragrance peaks at 20 minutes and gradually decreases over time.

2.5.2 Combustion quality

The easy ignition ability and burning characteristics were evaluated [Fig 2.5]. The samples were irradiated in a controlled chamber. The burning time was measured with a stopwatch, and the flame uniformity was noted. Observations were performed to ensure perfect combustion without leaving behind much smoke.

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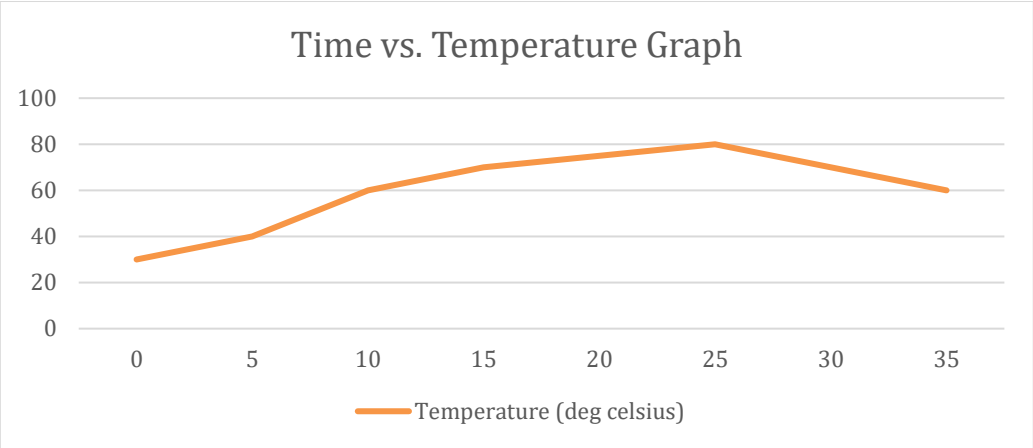

Figure No. 2.5 - The figure illustrates a time-temperature graph depicting that the temperature rises steadily due to combustion, peaking at 80 degrees Celsius at 25 minutes, and then begins to decline.

2.5.3 Smoke composition

The smoke emitted was qualitatively analyzed to guarantee that no carbonaceous emissions resulted. A smoke analyzer was used to measure particulate emissions and confirm the absence of carbonaceous material. The smoke was collected in a closed chamber for a qualitative analysis of visible soot and odor. The eco-friendly nature of the products was established by correlating their respective emissions with conventional products [Fig 2.6].

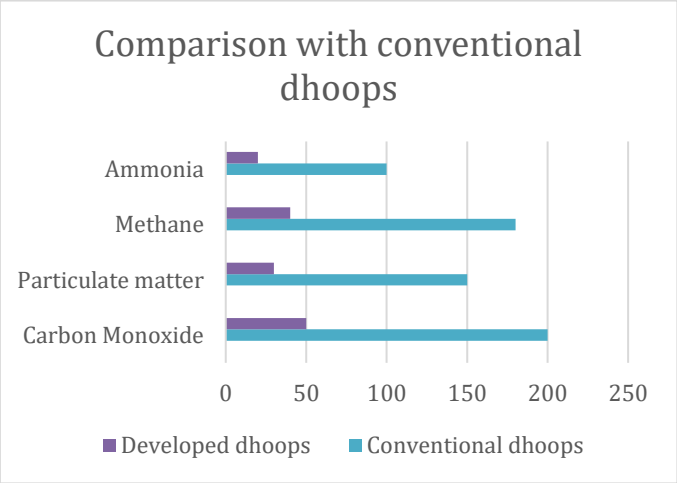

Figure No. 2.6 - The figure illustrates a bar graph comparing emissions (in ppm) of conventional vs. developed dhoop sticks, concluding that the developed dhoop sticks have significantly lower emissions across all categories.

2.5.4 Structural Integrity

The samples were placed in a humidity-controlled chamber for 48 hours at 70% and 90% relative humidity. Any changes in dimensions, cracks, or deformations were measured. This helped to understand the durability of the dhoop sticks while subjected to handling and storage conditions [Table 2.1].

Condition	Wax Based dhoops	Natural gum-based Dhoops	Comments
High humidity	Strong	Moderate	Slight softening in gum-based.
High temperature	Strong	Strong	Stable performance for both types.
Low temperature	Moderate	Strong	Wax tends to crack slightly
Normal conditions	Strong	Strong	No issues observed

Table No. 2.1 - The table illustrates the structural stability under varying conditions, depicting that the gum-based formulations maintain stability in varying conditions but are slightly affected by high humidity.

2.5.5 Ash Residue

The quantity and character of the ash residue [Fig 2.7] were checked to be minimum and fine in nature, signifying complete combustion. The ash residue was collected and weighed using a high-precision balance. The ash structure was examined under a microscope for fineness and consistency.

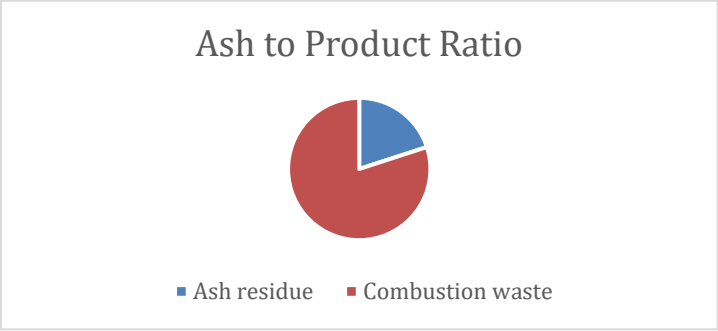

Figure No. 2.7 - This illustrates a pie chart comparing the ash-to-product ratio concluding that the developed dhoop sticks produce significantly less ash residue, highlighting eco-friendliness

2.5.6 Consumer feedback

The samples were distributed to 20 customers for assessment. Comments on flavor, burning characteristics, and overall acceptance were gathered and evaluated.[Fig 2.8]

Figure No. 2.8 - This figure illustrates the word cloud that was generated to visualize the most frequently used terms from the feedback.

3. Results and Discussion

3.1 Product development

An important part of this project was formulating the dhoop sticks. Various ingredient combinations were investigated to achieve optimum burning time, binding properties, and physical stability. The following summarizes some representative formulations, their rationale, and the corresponding observations.

Formulation 1:

Component Name	Proportion (%)	Quantity (gm)
Flower powder	55	5.5
Wax	17	1.7
Coconut oil	8	8
Camphor	10	1
Charcoal	10	1

Table No. 3.1 - This table illustrates the composition of formulation 1.

The formulation was solid but not porous. This hindered effective heat transfer during combustion. Coconut oil, a combustible liquid, was thus added to enhance heat transfer and burning efficiency [Table 3.1]. Therefore, the burning time was significantly improved, but the physical state needed further optimization to address the lack of cohesiveness.

To improve both the burning time and binding properties, the addition of wood powder into the formulation was incorporated [Table 3.2].

Formulation 2:

Component Name	Proportion (%)	Quantity (gm)
Flower powder	50	2.5
Wax	10	0.5
Coconut oil	11	0.4
Camphor	10	0.5
Charcoal	10	0.6
Wood powder	50	2.5
Essential oil	11	0.25

Table No. 3.2 - This table illustrates the composition of formulation 2.

This formulation exhibited good burning time and significantly improved binding, resulting in a more stable and cohesive physical state.

Wood powder was replaced with cow dung powder as a binder to enhance the binding properties of the formulation [Table 3.3]. Essential oils added fragrance and supported the cohesiveness of the components.

Formulation 3:

Component Name	Proportion (%)	Quantity (gm)
Flower powder	50	2.5
Wax	10	0.5
Coconut oil	11	0.4

Camphor	10	0.5
Charcoal	10	0.6
Cow-dung powder	50	2.5
Essential oil	11	0.25

Table No. 3.3 - This table illustrates the composition of formulation 3.

This formulation exhibited excellent binding, thus creating a stable physical state without any structural cracking. Burning time was optimal, and fragrance diffusion was consistent. By emphasizing cow dung powder's role as an eco-friendly and effective binding agent, this adjustment aligns well with the sustainability goals of our project.

3.1.1 Binder Optimization

This section describes the systematic approach to finding a sustainable binder and demonstrates our iterative process in refining the formulation.

The use of wax as a binder was reconsidered to align with the sustainability goals of the project. Multiple natural alternatives were explored to replace wax while maintaining or enhancing the binding and burning properties. To replace wax with a more environmentally friendly binder, liquid Karanja oil was used as a binder, and steric acid [Table 3.4] was added to mimic the binding of wax. Steric acid is a non-reactive fatty acid found in plant and animal fat. It has a melting point of 69.3 degrees Celsius. However, this attempt led the formulation to be weakly cohesive as the percentage of solids in the recipe was much higher than that of the binders. Decreasing the percentage of solids [Table 3.5] and subsequently increasing the binder percentage was only partially successful in improving the physical state of the dhoop sticks. Epoxy Soyabean oil was used as an alternate binder [Table 3.6], which enhanced the binding to some extent but was still not strong enough to ensure the structure. The binding strength was further improved by using a mixture of palm oil and Mahua crude oil [Table 3.7]. The binding was much improved than that of earlier experiments. However, substituting coconut oil led to reduced burning properties. The binding was not as strong as that imparted by wax. The importance of wax to obtain good binding without affecting burning properties was made clear by this study. All the natural replacements had improved in some ways over wax but had not equated it in any of the characteristics, especially bond strength and combustion efficiency. Results would emphasize the search for a natural binder that, in its use, would both achieve high binding strength and burn efficiently

Formulation 4:

Component Name	Proportion (%)	Quantity (gm)
Flower powder	60	6
Cow-dung powder	10	1
Steric acid	10	1
Camphor	5	0.5
Karanja oil	5	0.5
Essential oil	7	0.7
Charcoal	3	0.3

Table No. 3.4 - This table illustrates the composition of formulation 4.

Formulation 5:

Component Name	Proportion (%)	Quantity (gm)
Flower powder	40	4
Cow-dung powder	10	1

Steric acid	15	1.5
Camphor	5	0.5
Karanja oil	20	2
Essential oil	7	0.7
Charcoal	3	0.3

Table No. 3.5 - This table illustrates the composition of formulation 5.

Formulation 6:

Component Name	Proportion (%)	Quantity (gm)
Flower powder	40	4
Cow-dung powder	10	1
Steric acid	15	1.5
Camphor	5	0.5
Epoxy Soyabean oil	20	2
Essential oil	7	0.7
Charcoal	3	0.3

Table No. 3.6 - This table illustrates the composition of formulation 6.

Formulation 7:

Component Name	Proportion (%)	Quantity (gm)
Flower powder	40	2
Cow-dung powder	20	1
Steric acid	10	0.5
Essential oil	5	0.25
Crude Mahua oil	10	0.56
Palm oil	10	0.5
Camphor	5	0.25

Table No. 3.7 - This table illustrates the composition of formulation 7.

3.1.2 Natural Gum-based Binder Optimization

This section depicts the implementation of sustainable solutions. It also demonstrates how natural gums resolved multiple issues at once, highlighting the innovation in our formulation process.

In order to overcome the drawbacks of the previous formulations and in accordance with the sustainability objectives, natural gums, and plant-based materials were considered as potential binders. These materials were supposed to enhance the binding strength along with the burning efficiency and remove the extra additives such as camphor and charcoal.

Xanthan Gum, Babool Gum, gandh Biroza and Gum benzoin, like natural gums, were used for their excellent binding properties [Table 3.8]. Gum Benzoin was especially useful because it performed the role of a flux along with promoting the burning. Gugal, Jigat powder, and Gond Powder [Table 3.9] helped in strengthening the material along with imparting a pleasant natural scent; there was minimal need for synthetic or extra fragrance ingredients.

Natural gums definitely increased the strength of binding; it formed a solidly bonded material [Table 3.10]. The physical stability exhibited by the formulations did not crack or break into small parts. Gum Benzion and Gugal were not only used as binders but also enhanced the burning efficiency by ensuring a steady combustion process. The removal of charcoal and camphor streamlined the composition without losing the burn rate. The natural fragrance provided by Gum benzoin, gugal, and Jigat powder negated the requirement of

artificial or external fragrance enhancers. This gave it a unique, sweet fragrance, which made it even more appealing for use in dhoop sticks. This resolved the challenge of binder optimization sustainably and effectively. These materials met the requirements of binding, besides improving the burning efficiency and fragrance, which made the formulation more eco-friendly and streamlined.

Formulation 8:

Component Name	Proportion (%)	Quantity (gm)
Flower powder	40	2
Cow-dung powder	15	0.75
Palm oil	5	0.25
Mahua oil	5	0.25
Gugal	5	0.25
Gum Benzoin	5	0.25
Steric acid	25	1.25

Table No. 3.8 - This table illustrates the composition of formulation 8.

Formulation 9:

Component Name	Proportion (%)	Quantity (gm)
Flower powder	52	5.2
Cow-dung powder	10	1
Palm oil	5	0.5
Mahua oil	5	0.5
Gugal	5	0.5
Gum Benzoin	5	0.5
Steric acid	5	0.5
Jigat powder	13	1.3

Table No. 3.9 - This table illustrates the composition of formulation 9.

Formulation 10:

Component Name	Proportion (%)	Quantity (gm)
Flower powder	35	3.5
Cow-dung powder	5	0.5
Jigat	10	1
Gandh Biroza	10	1
Mahua crude oil	20	2
Palm oil	20	2

Table No. 3.10 -This table illustrates the composition of formulation 10.

3.2 Quality Assessments

The formulations were systematically tested based on the identified parameters: Burning efficiency, binding strength, fragrance, release, and eco-friendliness. The formulations were tested for Burning efficiency, focusing on the duration, consistency, and uniformity of burning. The burning test was carried out using standard conditions. The flame stability and smoke generation were observed.

It was observed that the formulations containing coconut oil and wax exhibited prolonged and consistent burning due to optimal calorific value. Substituting with oils like palm oil, mahua crude oil, and epoxy soybean oil initially decreased the burning efficiency. However, the addition of Gum Benzoin and Gugal improved the burning properties. The absence of camphor and charcoal decreased the smoke generation, and hence, the product is eco-friendly.

The cohesiveness and strength of the sticks against mechanical stress were tested using a drop test and a manual compression test. Wax-based formulations gave excellent binding strength with less cracking. Natural gums like Babool Gum, Xanthan Gum, and gond powder worked well as binders. They gave excellent binding strength and did not break easily.

Experiments with stearic acid and karanaja oil showed poor binding. The ratio of binder to solid was inappropriate, which caused structural instability. A sensory test was conducted by a group of testers to assess the strength and quality of the fragrance when burned. Consistency of fragrance was observed during the burning cycle. Formulation with essential oils along with Gum Benzoin and Gugal gave a long-lasting and natural fragrance. Using natural gums in place of camphor and charcoal was an eco-friendly option that did not compromise the fragrance. The powder formulations based on flowers gave a positive floral fragrance, which enhanced user acceptability.

The formulations were analyzed for carbonaceous emission and ash residue on complete burning. This helped to establish the environment's acceptability of the developed formulations. All of them had a low environmental impact with no detectable carbonaceous emissions. Products made from natural binders and essential oils produced very minimal ash residue, in line with sustainable practices. Temple flower waste usage reduced the waste burden while upholding circular economy principles.

3.3 Consumer Feedback

The consumer feedback questionnaire (Annexure) was distributed among participants to assess the quality, burning efficiency, and overall acceptance of the developed dhoop sticks. The compiled data is presented in Table 3.11, with a majority of users expressing positive responses toward fragrance, binding strength, and eco-friendliness."

3.3.1 Consumer Feedback Questionnaire

1) How would you rate the fragrance of the dhoop sticks?
   ☐ Excellent ☐ Good ☐ Average ☐ Poor
2) How long did the fragrance last after burning?
   ☐ <10 minutes ☐ 10–20 minutes ☐ 20–30 minutes ☐ >30 minutes
3) Did you experience any irritation or discomfort due to smoke?
   ☐ Yes ☐ No
4) How would you rate the binding strength and structural stability?
   ☐ Excellent ☐ Good ☐ Average ☐ Poor
5) Compared to conventional dhoop sticks, how do you find the developed product?
   ☐ Much Better ☐ Slightly Better ☐ Same ☐ Worse
6) Would you prefer to use these eco-friendly dhoop sticks regularly?
   ☐ Yes ☐ No
7) Any additional comments or suggestions? (open-ended response)

3.3.2 Compiling the responses

The response obtained was compiled into Table 3.11, and the feedbacks were assessed accordingly.

Assessment data	Excellent (%)	Good (%)	Average (%)	Poor (%)
Fragrance quality	60	30	10	0
Fragrance Retention	50	40	10	0
No smoke irritation	90	-	-	10
Binding Strength	70	20	10	0
Preference over conventional	80	15	5	0
Willingness to use	85	-	-	15

Table 3.11 - Compilation of the data received from the feedback of the customers

3.4 Environmental Impact

The conventional practice of temple flower waste disposal is highly unfavorable to the environment.[2]

1) *Water pollution:* When flowers are dumped in water bodies, they cause eutrophication, reduce oxygen levels, and negatively affect the aquatic ecosystem. [12]
2) *Land pollution:* Flowers dumped in open places decompose, which results in bad smells, attracts pests, and may even contaminate the soil.[13]
3) *Air pollution:* In some cases, the burning of flower waste causes air pollution.[14], [15]

We used organic materials like cow dung (as a binder), palm oil, and camphor in our composition. Our product contains the natural essential oils of white udh, sandalwood, and jasmine species. This project converts temple flower waste into dhoop sticks, thus offering several environmental benefits:

1) *Waste Reduction:* It significantly reduces the amount of flower waste that ends up in landfills or water bodies.
2) *Resource Conservation:* The waste is used as a product, and thus, resources are saved.
3) *Less Dependence on Synthetics:* No synthetic chemicals or fragrances are used, as in the case of commercial dhoop sticks.
4) *Cleaner Air:* It is organic in nature and has less smoke compared to the usual dhoop sticks, therefore causing less air pollution.
5) *Lower Carbon Footprint:* Due to the fact that it's organic waste, the carbon footprint in the production is lower compared to the traditional dhoop stick, potentially containing synthetic chemicals as well as being manufactured through a more energy-intensive process.

3.5 Challenges

The hardest thing was finding a strong, robust stick to work with. We tried different binders in hopes of fixing the problem. Some of the mixtures didn't burn all the way. We didn't face many problems concerning our project process. If we optimized some steps, most of the issues could be solved.

This research successfully describes the potential to turn waste temple flowers into non-pollutive dhoop sticks and incense sticks. While investigating various compositions and optimizing the natural adhesives using cow dung, gum benzoin, and gugal, the program successfully could carry out the task of rendering an environment-efficient method to dispose of waste. The usage of synthetics such as camphor and charcoal present in contemporary products combined with the usage of natural essential oils has managed to decrease the adverse impact to a considerable degree from this product.

These ready dhoop sticks will possess advantageous characteristics, such as a good burn, sweet fragrance, and less ash. However, the solution for temple floral waste management simultaneously aligns the production with circular economics as this production of waste will ultimately turn waste into valuable items, enhancing the possibilities based on its conclusion to open bigger opportunities with encouragement to similar action from local authorities in keeping it safe by doing sustainable acts within their environments.

4. Conclusion

This study is highly illustrative of the possibility of transforming temple flower waste into eco-friendly dhoop sticks. In studying numerous formulations and optimizing the usage of natural binders such as cow dung, gum benzoin, and gugal for the making of dhoop sticks, the initiative succeeded in providing a sustainable and efficient approach to waste

management. The exclusion of synthetic substances such as camphor and charcoal, along with the incorporation of natural essential oils, significantly reduces the environmental footprint of the final product.

The dhoop sticks created have desirable characteristics, such as steady burning, a pleasant aroma, and minimal ash. This project addresses the environmental concerns associated with temple flower waste while promoting a circular economy by converting waste into a valuable product. The findings of this research can be useful in scaling up similar initiatives and encouraging community participation in environmental conservation through innovative and sustainable practices.

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